ASSAM SCIENCE AND TECHNOLOGY UNIVERSITY
GUWAHATI

Course Structure and Syllabus
(From Academic Session 2018-19 onwards)

B.TECH

ELECTRONICS AND COMMUNICATION ENGINEERING/
ELECTRONICS AND TELECOMMUNICATION ENGINEERING

7th SEMESTER

ASSAM SCIENCE AND TECHNOLOGY UNIVERSITY
Guwahati
Course Structure

(From Academic Session 2018-19 onwards)
B.Tech 7th Semester: Electronics and Communication/Telecommunication Engineering

Semester VII/ B.TECH/ECE/ETE

Sl. No	Sub-Code	Subject	Hours per Week			Credit	Marks	
			L	T	P	C	CE	ESE
Theory								
1	ECE181701	Microwave Engineering	3	0	2	4	30	70
2	ECE181702	VLSI System Design	3	0	0	3	30	70
3	ECE1817PE2*	Program Elective-2	3	0	0	3	30	70
4	ECE1817OE2*	Open Elective-2	3	0	0	3	30	70
5	HS181704	Principles of Management	3	0	0	3	30	70
Practical								
1	ECE181712	VLSI System Design Lab	0	0	2	1	15	35
2	ECE181722	Project-1	0	0	6	3	50	50
3	SI181721	Internship-III (SAI - Industry)	0	0	0	2	-	200
TOTAL			**15**	**0**	**10**	**22**	**215**	**635**
Total Contact Hours per week : 25								
Total Credit: 22								

PROGRAM ELECTIVE – 2		
Sl.No.	**Subject Code**	**Subject**
1	ECE1817PE21	MEMS
2	ECE1817PE22	Nanoelectronics
3	ECE1817PE2*	Any other subject offered from time to time with the approval of the University

OPEN ELECTIVE – 2		
Sl.No.	**Subject Code**	**Subject**
1	ECE1817OE21	Image Processing
2	ECE1817OE22	Information Theory And Coding
3	ECE1817OE23	IoT
4	ECE1817OE2*	Any other subject offered from time to time with the approval of the University

Detail Syllabus:

Course Code	Course Title	Hours per week L-T-P	Credit C
ECE181701	**Microwave Engineering**	**3-0-2**	**4**

Course Outcomes:

At the end of the course, the students will be able to:

CO1: Apply electromagnetic theory in calculations of waveguide and transmission line parameters.
CO2: Make use of scattering matrix to different types of microwave devices.
CO3: Analyze various Passive and Active Microwave Devices.
CO4: Measure parameters at microwaves frequencies.
CO5: Design microwave systems for different practical application.

MODULE 1: Introduction to Microwaves
Introduction to Microwaves: History of Microwaves, Microwave Frequency bands;
Applications of Microwaves: Civil and Military, Medical, EMI/ EMC.

MODULE 2: Microwave Transmission lines and Waveguides
Review of transmission line theory, Co-axial cable, MIC lines, Characteristic impedance, Standing waves, VSWR and reflection coefficient, Smith chart, Stub matching, Quarter wave transformer. Rectangular and circular waveguides. Solution of wave equations. TE, TM and TEM modes. Dominant mode. Filed Patterns. Cut-off frequencies. Wave impedance. Power transmission. Waveguide resonators.

MODULE 3: Microwave Network Analysis
Network parameters for microwave circuits, Scattering Parameters.

MODULE 4: Passive and Active Microwave Devices
Microwave passive components: Directional Coupler, Power Divider, Magic Tee, Attenuator, Resonator.
Microwave active components: Diodes, Transistors, Oscillators, Mixers.
Microwave Semiconductor Devices: Gunn Diodes, IMPATT diodes, Schottky Barrier diodes, PIN diodes.
Microwave Tubes: Klystron, TWT, Magnetron.

MODULE 5: Microwave Measurements
Microwave Measurements- Power, Frequency and impedance measurement at microwave frequency, Network Analyzer and measurement of scattering parameters, Spectrum Analyzer and measurement of spectrum of a microwave signal, Noise at microwave frequency and measurement of noise figure. Measurement of Microwave antenna parameters.

MODULE 6: Microwave Design Principles
Impedance transformation, Impedance Matching, Microwave Filter Design, RF and Microwave Amplifier Design, Microwave Power Amplifier Design, Low Noise Amplifier Design, Microwave Mixer Design, Microwave Oscillator Design. Microwave Antennas- Antenna parameters, Antenna for ground based systems, Antennas for airborne and satellite borne systems, Planar Antennas.

Text/Reference Books:
1. R.E. Collins, Microwave Circuits, McGraw Hill
2. K.C. Gupta and I.J. Bahl, Microwave Circuits, Artech house

Course Code	Course Title	Hours per week L-T-P	Credit C
ECE181702	**VLSI System Design**	**3-0-0**	**3**

Course Outcomes:
At the end of the course, the students will be able to

CO1: Explain different processes involved in IC fabrication technique.
CO2: Design combinational and sequential circuits using different logic style.
CO3: Analyze the circuit parameters and understand their effects.
CO4: Design different arithmetic building blocks and semiconductor memory structures.
CO5: Evaluate fault modeling in VLSI design.

MODULE 1: IC Design and Fabrication Process

General overview of design hierarchy, layers of abstraction, integration density and Moore's law; MOSFET fabrication steps, Basic steps of BJT, p-MOS, n-MOS fabrication, CMOS p-well and n-well processes, Bi-CMOS fabrication process; Stick diagram and Layout of circuits using λ – based rules and color schemes of various layers.

MODULE 2: Design of MOSFFT Based Digital ICs

Basic Structure and operation of a MOSFET, MOS transistor threshold voltage, MOS switch, CMOS inverter and characteristics, MOSFET scaling - constant-voltage and constant-field scaling, Static CMOS Construction, Pass transistor logic, Transmission gates, Dynamic logic design.

MODULE 3: Circuit Characteristic and Performance Estimation

Logical effort & transistor sizing, parasitic delay in a logic gate, linear delay model, multi-stage logic networks, choosing of best number of stages. Power Dissipation in VLSI circuits (Static, dynamic).

MODULE 4: Subsystem Design

Design of arithmetic building blocks like adders (static, dynamic, Manchester carry-chain, look-ahead, carry bypass and pipelined adders) and multipliers (serial-parallel, Braun, Baugh-Wooley and systolic array multipliers); barrel and logarithmic shifters; designing semiconductor memory and array structures (SRAM and DRAM).

MODULE 5: Fault Modeling and Testing in VLSI Design

Defects in chip, common fault models, Fault simulation and testability measures, combinational circuit test pattern generation, sequential circuit testing and scan chains, built in self-test.

Text/Reference Books:

1. N.H.E. Weste and K. Eshraghian, Principles of CMOS VLSI Design - a System Perspective, 2nd ed., Pearson Education Asia, 2002
2. J.M. Rabaey, A. Chandrakasan and B. Nikolic, Digital Integrated Circuits- A Design Perspective, 2nd ed., PHI, 2003

3. S.M. Kang and Y. Leblevici, CMOS Digital Integrated Circuits Analysis and Design, 3rd ed., McGraw Hill, 2003
4. R. Jacob Baker, CMOS Circuit Design, Layout, and Simulation, IEEE Press, 1997.
5. S.M. Sze, VLSI Technology, Mc Graw Hill
6. Pucknell, Basic VLSI design, PHI
7. George W Zobrist , VLSI Fault Modeling and Testing Techniques

Course Code	Course Title	Hours per week L-T-P	Credit C
ECE1817PE21	**MEMS**	**3-0-0**	**3**

Course Outcomes:

At the end of the course the students will be able to

- **CO1:** Define MEMS for different applications.
- **CO2:** Identify structural and sacrificial materials for MEMS.
- **CO3:** Choose suitable fabrication steps for designing various MEMS devices.
- **CO4:** Select appropriate sensors for various applications.

MODULE 1: Introduction to MEMS

Introduction to MEMS and Microsystems, MEMS Classification, MEMS versus Microelectronics, Applications of MEMS in Various Industries, Some Examples of Microsensors, Microactuators, and Microsystems, Materials for MEMS, Laws of Scaling in miniaturization.

MODULE 2: Material Properties

MEMS materials, structural and sacrificial materials, properties of silicon, mechanical, electrical and thermal properties of materials, Basic modeling of elements in electrical and mechanical systems.

MODULE 3: Fabrication Methods

MEMS Fabrication Technologies, single crystal growth, micromaching, photolithography, microsterolithography, thin film deposition, impurity doping, diffusion, etching, bulk and surface micromaching, etch stop technique and microstructure, LIGA.

MODULE 4: Mechanical Sensors & Actuators

Stress and Strain, Hooke's Law. Stress and Strain of Beam Structures, Cantilever, Pressure sensors, Piezoresistance Effect, Piezoelectricity, Piezoresistive Sensor, capacitive sensors, Inductive sensors, MEMS inertial sensors, micromachined microaccelerometer for MEMS, Parallel-plate Actuator, piezoactuators.

MODULE 5: Magnetic Sensors

Magnetic material for MEMS, magnetic sensing and detection, mannetoresistive sensors, hall effect, magnetodiode, megnetotransitors, MEMS magnetic sensors, RF MEMS.

MODULE 6: Thermal Sensors

Temperature coefficient of resistance, Thermo-electricity, Thermocouples, Thermal and temperature sensors, heat pump, micromachined thermocouple probe, thermal flow sensors, shape memory alloy.

Text/Reference Books:

1. Analysis and Design Principles of MEMS Devices by Minhang Bao, ELSEVIER.
2. M. J. Usher, “Sensors and Transducers”, McMillian Hampshire.
3. N. P. Mahalik, “MEMS” Tata McGraq Hill.
4. R.S. Muller, Howe, Senturia and Smith, “Microsensors”, IEEE Press.
5. S. M. Sze, Semiconductor Sensors, Willy –Interscience Publications.

Course Code	Course Title	Hours per week L-T-P	Credit C
ECE1817PE22	**Nanoelectronics**	**3-0-0**	**3**

Course Outcomes:

At the end of the course the students will be able to:

- **CO1:** Compare the characteristics of microelectronic and nanoelectronic devices.
- **CO2:** Explain basic and advanced concepts of nano-electronic devices, sensors and transducers and their applications in nanotechnology.
- **CO3:** Identify different fabrication and characterization techniques for nano-scale devices.
- **CO4:** Distinguish between diffusive and ballistic transport of electrons in nanostructures.
- **CO5:** Assess the characteristics of selected Nanoelectronic devices.

MODULE 1: Introduction

Introduction to nanotechnology and nanoelectronics, Impacts, Limitations of conventional microelectronics. Transition from classical electronics to nanoelectronics

MODULE 2: Basics of Quantum Mechanics

Schrodinger equation, Density of States. Particle in a box Concepts, Degeneracy. Band Theory of Solids. Kronig-Penny Model. Brillouin Zones

MODULE 3: Nanoelectronic Materials

Semiconductors, Crystal lattices, Bonding in crystals, Electron energy bands, Electrons in low-dimensional structures - Electrons in quantum wells - Electrons in quantum wires - Electrons in quantum dots, Semiconductor heterostructures, Inorganic-organic heterostructures. Carbon nanomaterials

MODULE 4: Growth and Fabrication of Nanostructures

Introduction to methods of fabrication of nanomaterials-different approaches. fabrication of nano-layers -Physical Vapor Deposition, Chemical Vapor Deposition, Epitaxy, Molecular Beam Epitaxy, Ion Implantation, Formation of Silicon Dioxide. Fabrication of nanoparticle- grinding with iron balls, laser ablation, reduction methods, sol gel, hydrothermal, self-assembly, precipitation of quantum dots

MODULE 5: Characterization of Nanostructures

Introduction to characterization tools of nanostructures - -principle of operation of STM, AFM, SEM, TEM, XRD, PL & UV instruments

MODULE 6: Electron Transport in Nanostructures

Introduction to electron transport phenomenon, Diffusive and Ballistic transport of electrons, Time and length scales of the electrons in solids, Statistics of the electrons in solids and nanostructures, Density of states of electrons in nanostructures, Electron transport in nanostructures

MODULE 7: Nanoelectronic Devices

Resonant-tunneling diodes, Field-effect transistors, Single-electron-transfer devices

Text/Reference Books:

1. George W. Hanson, "Fundamentals of Nanoelectronics", PHI
2. V. Mitin, V. Kochelap, and M. Stroscio "Introduction to Nanoelectronics: Science, Nanotechnology, Engineering, and Applications", Cambridge University Press, 2008

Course Code	Course Title	Hours per week L-T-P	Credit C
ECE1817OE21	**Image Processing**	**3-0-0**	**3**

Course Outcomes:

At the end of the course the students will be able to:

CO1: Explain the fundamental concepts of a digital image processing system

CO2: Analyse images in different domains using various transforms

CO3: Evaluate the techniques for image enhancement and image restoration

CO4: Understand the need for image compression and interpret image compression standards

CO5: Explain colour image processing

MODULE 1: Introduction

Scope and application of digital image processing. Image acquisition and display. Mathematical preliminaries. Human visual perception.

MODULE 2: Image Transforms

2D-Fourier Transforms. 2D DFT, KLT, 2D DCT, Haar transform.

MODULE 3: Image Enhancement

Histogram processing. Spatial Filtering. Frequency Domain Filtering.

MODULE 4: Image Restoration

Degradation Model. Inverse Filtering. Wiener Filtering.

MODULE 5: Edge Detection and Segmentation

Edge detection. Line detection. Segmentation. Texture

MODULE 6: Image Compression

Lossy Compression. Loss-less compression. Run-length and Huffman Coding. Transform Coding. Image Compression Standards

MODULE 7: Color Image Processing

Color models, Color Image Processing

Text/Reference Books:

1. R. C. Gonzalez & R. E. Woods,"*Digital Image Processing*", Addison Wesley, 1993.
2. A. K. Jain, "*Fundamentals of Digital Image Processing*", PHI
3. K. R. Castleman, "*Digital Image Processing*", PHI 1996
4. W. K. Pratt, "*Digital Image Processing*", John Wiley Interscience, 1991

Course Code	Course Title	Hours per week L-T-P	Credit C
ECE1817OE22	**Information Theory and Coding**	**3-0-0**	**3**

Course Outcomes:

At the end of the course, the students will be able to:

CO1: Explain different types of entropy of a communication channel.

CO2: Analyse the information-carrying capacity of communication channels.

CO3: Choose suitable source coding techniques to improve the efficiency of information transmission.

CO4: Apply different codes for error detection and correction.

MODULE 1:

Concept of amount of information, information units Entropy: marginal, conditional, joint and relative entropies, relation among entropies Mutual information, information rate, channel capacity, redundancy and efficiency of channels Discrete channels – Symmetric channels, Binary Symmetric Channel, Binary Erasure Channel, Noise-Free Channel, Channel with independent I/O, Cascaded channels, repetition of symbols, Binary asymmetric channel, Shannon theorem

MODULE 2:

Source coding – Encoding techniques, Purpose of encoding, Instantaneous codes, Construction of instantaneous codes, Kraft's inequality, Coding efficiency and redundancy, Source coding theorem. Construction of basic source codes – Shannon Fano coding, Shannon Fano Elias coding, Huffman coding, Minimum variance Huffman coding, Adaptive Huffman coding, Arithmetic coding, Dictionary coding – LZ77, LZ78, LZW, ZIP coding Channel coding, Channel coding theorem for DMC

MODULE 3:

Codes for error detection and correction – Parity check coding, Linear block codes, Error detecting and correcting capabilities, Generator and Parity check matrices, Standard array and Syndrome decoding, Hamming codes Cyclic codes – Generator polynomial, Generator and Parity check matrices, Encoding of cyclic codes, Syndrome computation and error detection, Decoding of cyclic codes, BCH codes, RS codes, Burst error correction

MODULE 4:

Convolutional codes – Encoding and State, Tree and Trellis diagrams, Maximum likelihood decoding of convolutional codes -Viterbi algorithm, Sequential decoding -Stack algorithm. Interleaving techniques – Block and convolutional interleaving, Coding and interleaving applied to CD digital audio system - CIRC encoding and decoding, interpolation and muting. ARQ – Types of ARQ, Performance of ARQ, Probability of error and throughput.

Text Books:

1. Elements of Information Theory by Thomas Cover, Joy Thomas
2. Channel Codes: Classical and Modern by William Ryan, Shu Lin

References:

1. Information Theory and Reliable Communication by Robert Gallager

Course Code	Course Title	Hours per week L-T-P	Credit C
ECE1817OE23	**IoT**	**3-0-0**	**3**

Course Outcomes:

At the end of the course the students will be able to:

CO1: Explain IoT structure (device, data cloud) and application areas.

CO2: Analyse IoT sensors and technological challenges faced by IoT devices, with a focus on wireless, energy, power, and sensing modules.

CO3: Compare different communication protocols for IoT applications.

CO4: Interpret data for various IoT applications.

MODULE 1: IoT Introduction and Fundamentals

Deciphering the term IoT; Applications where IoT can be deployed; Benefits/challenges of deploying an IoT; IoT components: Sensors, front-end electronics (amplifiers, filtering, digitization), digital signal processing, data transmission, choice of channel (wired/wireless), back-end data analysis; Understanding packaging and power constraints for IoT implementation

MODULE 2: Signals, Sensors, Actuators, Interfaces

Sensors: types, signal types, shape and strength; Sensor non-idealities: Sensitivity and offset drift, noise, minimum detectable signal, nonlinearity; Read-out circuits: Instrumentation-amplifier, SNR definition, noise-bandwidth-power tradeoff; Circuit component mismatch and mitigation techniques (calibration, chopping, autozeroing etc.); Power/energy considerations; Basic signal processing (filtering, quantization, computation, storage)

MODULE 3: Networking and Cloud Computing in IoT

Review of Communication Networks, Challenges in Networking of IoT Nodes, range, bandwidth; Machine-to-Machine (M2M) and IoT Technology Fundamentals, Medium Access Control (MAC) Protocols for M2M Communications 2 of 2; Standards for the IoT; Basics of 5G Cellular Networks and 5G IoT Communications, Low-Power Wide Area Networks (LPWAN); Wireless communication for IoT: channel models, power budgets, data rates; IoT Security and Privacy, MQTT Protocol, Publisher and Subscriber Model; Cloud computing platform (open source) and local setup of such environment;

Embedded software relevant to microcontroller and IoT platforms (enterprise or consumer), user interfaces

MODULE 4: Data Analysis for IoT applications

Statistics relevant to large data; Linear regression; Basics of clustering, classification

Text/Reference Books:

1. Vijay Madisetti and Arshdeep Bahga, “Internet of Things (A Hands-on-Approach)”, 1st Edition, VPT, 2014
2. Francis daCosta, “Rethinking the Internet of Things: A Scalable Approach to Connecting Everything”, 1st Edition, Apress Publications, 2013
3. Cuno Pfister, Getting Started with the Internet of Things, O"Reilly Media, 2011, ISBN: 978-1-4493-9357-1

Course Code	Course Title	Hours per week L-T-P	Credit C
HS181704	**Principles of Management**	**3-0-0**	**3**

MODULE 1: Introduction **(6 Lecture)**
Definition and meaning of management, Characteristics of management, importance of management, functions of management-planning, organising, directing, staffing, coordination and controlling etc. principles of management, Difference between administration and management

MODULE 2: Financial Management **(6 Lecture)**
Definition and management of financial planning, importance and characteristics of sound financial plan, concepts of capital- fixed capital and working capital, source of finance, fund flow statement.

MODULE 3: Marginal Costing **(6 Lecture)**
Definition and meaning of marginal costing, advantages, marginal cost equation, contribution, profit-volume ratio, break even analysis, margin of safety.

MODULE 4: Cost Accounting **(6 Lecture)**
Cost Accounting- Concept and benefit, elements of cost, preparation of cost sheet with adjustment of raw materials, work-in-progress and finished goods.

MODULE 5: Capitalisation **(4 Lecture)**
Definition and meaning of capitalisation, over and under capitalisation.

MODULE 6: Motivation **(6 Lecture)**
Introductory observation, definition of motivation, motivational technique, features of sound motivational system.

MODULE 7: Leadership **(6 Lecture)**
Concept of leadership, principles of leadership, functions of leadership, qualities of leadership, different styles of leadership**.**

Textbooks/Reference Books:

1. Principle of Business Management: RK Sharma, Shashi K.Gupta
2. Business Organisation and Management: SS Sarkar, RK Sharma, Shashi K.Gupta
3. Industrial Organisation and Management: SK Basu, KC Sahu, B Rajviv
4. Principles of Management by Dr. A. K. Bora: Chandra Prakash, Guwahati.
5. Management Accounting: RK Sharma, Shashi K Gupta
6. Cost Accounting: SP Jain, K I Narang
7. Cost Accounting, RSN Pillai, V Bhagawati
8. Principles of Management: RN Gupta
9. Principles of Management: RSN Pillai, S. Kala
10. Principles of Management: Dipak Kumar Bhattacharjee

Course Code	Course Title	Hours per week L-T-P	Credit C
ECE181712	**VLSI System Design Lab**	**0-0-2**	**1**

LIST OF EXPERIMENTS:

Exp1. Simulate the IV characteristics of NMOS transistor

Exp2. a) Simulate the schematic of the CMOS inverter to verify its VTC curve, perform transient analysis and estimate power, delay for the circuit.
b) Perform the physical verification for the layout of the CMOS inverter by DRC rule.

Exp3. a) Design a CMOS NAND gate and verify its functioning.
b) Perform the physical verification for the layout of CMOS NAND gate by DRC rule.

Exp4. a) Design a CMOS NOR gate and verify its functioning.
b) Perform the physical verification for the layout of CMOS NOR gate by DRC rule.

Exp 5. Design a transmission gate logic using the schematic entry tool and verify its functioning.

Exp 6. Design a pass transistor logic using the schematic entry tool and verify its functioning.

Exp 7. Design a MOS based SRAM cell and verify its characteristic.

Course Code	Course Title	Hours per week L-T-P	Credit C
ECE181722	**Project-1**	**0-0-6**	**3**
GUIDELINES WILL BE ISSUED BY THE UNIVERSITY FROM TIME TO TIME			

Course Code	Course Title	Hours per week L-T-P	Credit C
SI181721	**Internship-III (SAI - Industry)**	**0-0-0**	**2**
GUIDELINES WILL BE ISSUED BY THE UNIVERSITY FROM TIME TO TIME			

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